

LIBREVILLE INTL, Gabon

WMO: 645000

Lat: 0.459N

Lon: 9.412E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	71.6	73.0	66.3	97.1	75.2	67.7	101.8	75.7	16.2	76.8	14.5	76.9	6.0	80	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.4	89.2	81.2	88.1	80.8	87.6	80.6	82.8	86.4	82.1	85.7	81.5	85.2	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.2	167.7	85.8	80.9	160.9	84.6	80.7	159.5	84.4	47.0	86.9	46.2	86.7	45.5	85.3	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	13.4	12.1	DB	69.3	91.3	1.6	2.5	68.1	93.0	67.2	94.5	66.3	95.9	65.1	97.7	
			WB	67.2	84.7	1.4	1.7	66.2	85.9	65.4	86.9	64.6	87.8	63.6	89.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.1	81.1	81.7	81.9	81.7	81.3	79.3	77.5	77.8	79.0	79.5	79.8	80.6
	DBStd	2.17	1.62	1.92	1.72	1.78	1.64	1.57	1.57	1.37	1.37	1.48	1.59	1.60
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10980	966	887	987	950	971	878	851	863	871	915	893	948
	CDD65	5505	501	467	522	500	506	428	386	398	421	450	443	483
	CDH74	51835	5361	5260	5865	5427	5321	3580	2336	2483	3297	3908	4083	4914
	CDH80	12498	1540	1789	2100	1818	1504	447	150	171	355	586	816	1222
Wind	WSAvg	7.1	5.9	6.1	6.2	6.2	7.0	8.1	8.3	8.5	8.7	7.6	6.6	6.2
Precipitation	PrecAvg	109.0	10.3	9.8	14.1	12.7	9.7	0.7	0.1	0.3	4.6	16.7	19.5	12.0
	PrecMax	135.0	25.4	18.7	25.4	22.0	25.0	5.1	0.4	1.5	18.1	29.8	31.5	25.3
	PrecMin	77.8	3.1	3.1	3.1	0.4	1.0	0.0	0.0	0.0	0.1	0.0	9.7	2.3
	PrecStd	17.5	4.8	3.7	5.4	4.0	6.1	1.0	0.1	0.4	4.5	6.5	6.1	5.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	89.7	89.9	89.8	88.6	86.0	84.3	84.3	84.9	86.2	87.0	87.7
		MCWB	81.3	81.4	81.4	81.3	81.0	78.8	76.6	76.5	77.5	79.3	80.2	80.9
	2%	DB	87.5	88.2	89.2	88.4	87.7	84.4	82.7	82.7	84.1	84.6	86.0	86.3
		MCWB	80.9	81.0	81.1	80.9	80.7	77.8	75.7	75.5	77.3	78.6	79.8	80.1
	5%	DB	86.3	87.7	88.0	87.8	86.3	83.5	81.7	82.0	82.8	84.2	84.6	85.8
		MCWB	80.2	80.8	80.7	80.7	79.9	77.3	75.1	75.1	76.6	78.4	79.0	79.8
	10%	DB	85.7	86.4	87.4	86.5	85.7	82.5	80.6	80.7	82.3	82.8	84.0	84.5
		MCWB	79.9	80.1	80.4	80.0	79.6	76.7	74.4	74.4	76.4	77.7	78.7	79.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.1	83.4	83.7	83.6	83.1	80.1	78.7	78.3	79.8	81.4	81.8	82.5
		MCDB	86.5	87.2	87.8	87.4	86.7	83.3	81.6	81.7	82.5	84.0	84.7	85.4
	2%	WB	81.8	82.4	82.6	82.3	81.8	79.1	77.4	77.3	78.9	79.9	80.8	81.3
		MCDB	85.1	86.1	86.6	86.1	85.3	82.5	80.6	80.6	81.8	82.7	83.7	84.4
	5%	WB	81.0	81.6	81.7	81.4	80.9	78.3	76.5	76.4	78.2	79.1	80.0	80.6
		MCDB	84.4	85.5	85.9	85.5	84.6	81.8	79.9	79.8	81.1	82.1	83.0	83.7
	10%	WB	80.4	80.9	80.9	80.7	80.1	77.7	75.8	75.8	77.5	78.6	79.2	79.9
		MCDB	83.9	84.9	85.3	85.0	83.9	81.3	79.3	79.2	80.5	81.6	82.3	83.2
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.5	11.4	10.9	9.4	7.4	7.4	7.5	7.5	8.1	9.5	9.3
		MCDBR	10.2	11.0	12.2	11.8	10.4	8.3	8.4	8.4	8.2	8.9	10.3	9.9
		MCWBR	5.7	5.8	6.5	6.4	5.7	4.7	4.9	4.5	4.5	4.9	5.9	5.6
	5% WB	MCDBR	10.0	10.9	11.7	11.3	10.3	8.0	7.9	8.0	7.9	8.5	10.0	9.7
		MCWBR	6.3	6.6	7.1	6.8	6.1	5.1	5.2	4.9	5.1	5.3	6.3	6.1
Clear-Sky Solar Irradiance	taub	0.694	0.669	0.601	0.483	0.501	0.583	0.622	0.590	0.490	0.456	0.515	0.659	
	taud	1.729	1.787	1.958	2.314	2.241	1.970	1.867	1.952	2.253	2.374	2.198	1.798	
	Ebn at Noon	216	225	240	263	251	226	219	233	264	276	259	222	
	Edh at Noon	77	74	62	42	44	56	63	60	45	40	48	71	
All-Sky Solar Radiation	RadAvg	1505	1491	1556	1492	1387	1224	1227	1218	1272	1318	1328	1404	
	RadStd	84	79	59	53	71	89	62	65	81	68	46	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page